FitBuddy — Phase 5: Project Development

Implementation checklist based on the supplied technical specification

Purpose

• Prepared from the supplied FitBuddy project document. This document is a submission-ready phase artifact/template; it does not claim that implementation, testing, screenshots, or deployment have been completed unless you add evidence from your actual project.

Environment Setup

Create a Python virtual environment, install the required dependencies, configure the Gemini API key through an environment variable/.env approach, and run the FastAPI server with Uvicorn.

Recommended Source Structure

• app/ — Python application files.

• app/main.py — FastAPI application entry point (use the actual filename/structure in your implementation).

• app/routes.py — route handlers.

• app/gemini_generator.py — workout generation.

• app/gemini_flash_generator.py — nutrition/recovery tip.

• app/updated_plan.py — feedback-based update.

• app/database.py — SQLite/SQLAlchemy persistence.

• templates/index.html — input page.

• templates/result.html — result/feedback page.

• templates/all_users.html — admin page.

• static/images/ — image assets where used.

• requirements.txt — dependencies.

• .env — local secrets; do not commit secrets.

Development Tasks

• Implement the user input form.

• Implement workout-plan generation through the configured Gemini model.

• Implement nutrition/recovery tip generation.

• Implement feedback-based plan updating.

• Implement SQLite/SQLAlchemy storage.

• Implement FastAPI routes.

• Implement Jinja2 dynamic rendering.

• Implement admin view.

• Connect all components and run the application locally.

Important Implementation Evidence

For the final GitHub submission, replace this checklist with actual evidence: screenshots of the running app, relevant source files, commit history, and a short README explaining how to install and run the project.

Security Note

Never upload a real Gemini API key to a public GitHub repository. Use an environment variable or .env file locally and add .env to .gitignore.